

LM06 Capital Structure

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1. Introduction

This learning module covers:

- Calculating and interpreting the weighted-average cost of capital (WACC) for a company
- Factors affecting capital structure
- Modigliani–Miller propositions regarding capital structure
- Optimal and target capital structure

2. The Cost of Capital

Cost of capital is the rate of return that the suppliers of capital require as compensation for their contribution of capital. Assume a company decides to build a steel plant and needs money or capital for it. Investors such as bondholders or equity holders will lend this capital to the company. Suppliers of capital will be motivated to part with their money for a certain period of time if the money invested can earn a greater return than it would earn elsewhere. In short, investors will invest if the return (IRR) is greater than the cost of capital.

A company has access to several sources of capital such as issuing equity, debt, or instruments that share characteristics of both debt and equity. Each source becomes a component of the company's funding and has a specific cost associated with it called the **component cost of capital**.

The cost of capital is the rate of return expected by investors for average-risk investment in a company. Investors will demand a higher rate of return for higher-than-average-risk investments. Similarly, investors will demand a lower rate of return for lower-than-average-risk investments.

One way of calculating this cost is to determine the **weighted average cost of capital (WACC)**, which is also called the **marginal cost of capital**. It is called marginal because it is the additional or incremental cost a company incurs to issue additional debt or equity.

Three common sources of capital are common shares, preferred shares, and debt. WACC is the cost of each component of capital in the proportion they are used in the company.

$$\text{WACC} = w_d r_d (1 - t) + w_p r_p + w_e r_e$$

where:

w_d = proportion of debt that the company uses when it raises new funds

r_d = before-tax marginal cost of debt

t = company's marginal tax rate

w_p = proportion of preferred stock the company uses when it raises new funds

r_p = marginal cost of preferred stock

w_e = proportion of equity that the company uses when it raises new funds

r_e = the marginal cost of equity

The weights are the proportions of the various sources of capital that the company uses. The weights should represent the company's **target capital structure** and not the current capital structure. For example, suppose that current capital structure of a company is 33.3% debt, 33.3% preferred stock and 33.3% common stock. To fund a new project, the company plans to issue more debt and its capital structure will change to 50% debt, 25% preferred stock, and 25% common stock. WACC calculations should be based on these new weights, i.e., the target weights.

Example

IFT has the following capital structure: 30 percent debt, 10 percent preferred stock, and 60 percent equity. IFT wants to maintain these weights as it raises additional capital. Interest expense is tax deductible. The before-tax cost of debt is 8 percent, cost of preferred stock is 10 percent, and cost of equity is 15 percent. If the marginal tax rate is 40 percent, what is the WACC?

Solution:

$$\text{WACC} = (0.3) (0.08) (1 - 0.4) + (0.1) (0.1) + (0.6) (0.15) = 11.44 \text{ percent}$$

Note: Before-tax cost of debt is given. Do not forget to calculate the after-tax cost.

Example

Machiavelli Co. has an after-tax cost of debt capital of 4 percent, a cost of preferred stock of 8 percent, a cost of equity capital of 10 percent, and a weighted average cost of capital of 7 percent. Machiavelli Co. intends to maintain its current capital structure as it raises additional capital. In making its capital budgeting decisions for the average risk project, what is the relevant cost of capital?

Solution:

The relevant cost of capital is 7%. The WACC using weights derived from the current capital structure is the best estimate of the cost of capital for the average risk project of a company.

3. Factors Affecting Capital Structure

Capital structure refers to the specific mix of debt and equity a company uses to finance its assets and operations. Issuers desire a capital structure that minimizes their weighted-average cost of capital and generally matches the duration of their assets.

Determinants of the Amount and Type of Financing Needed

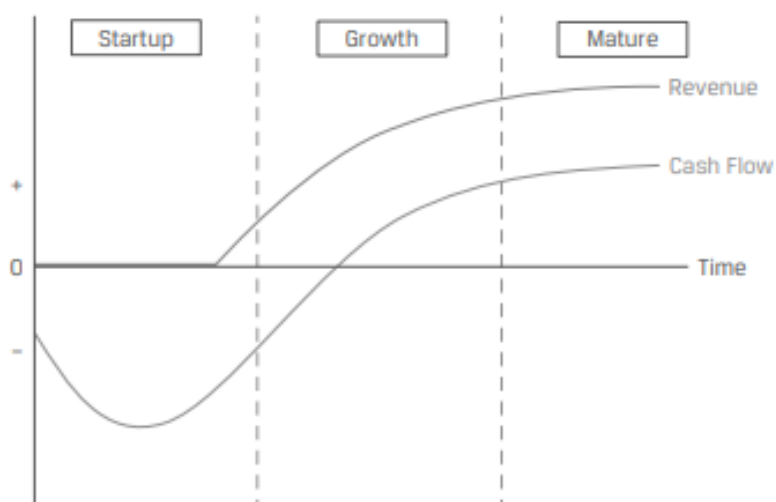
The total amount and type of financing needed usually depends on the issuer's business model and its position in the corporate life cycle.

Corporate Life Cycle

A company's life cycle stage influences its cash flow characteristics, its ability to support

debt; and is therefore a primary factor in determining capital structure. Any capital that is not sourced through borrowing must come from equity.

As companies mature and move from start-up, through growth, to maturity, their business risk typically declines, and their operating cash flows turn positive and become more predictable. This allows for greater use of leverage at more attractive terms. This is illustrated in Exhibit 2 from the curriculum.



Feature / Phase	Startup	Growth	Mature
Revenue Growth	Initial	Rising	Slowing
Free Cash Flow	Negative	Rising	Stable
Business Risk	High	Medium	Low
Debt Availability	Little or None	Rising	High
Debt Type	Convertible	Secured	Unsecured

Start-Ups

- In this stage a company's revenues are close to zero and a lot of investment is required to move from the prototype stage to commercial production.
- Therefore, cash flow is usually negative.
- The risk of business failure is high.
- The company typically raises capital through equity rather than debt.
- Equity is generally sourced through private markets (venture capital) rather than public markets.
- Debt is generally not available or is very expensive. It is usually a negligible component of the capital structure.
- Some start-ups may be able to raise convertible debt.

Growth Businesses

- In this stage a company typically experiences high revenue growth but investment is needed to achieve this growth.
- Therefore, cash flow may be negative but it is likely to be improving and becoming more predictable.
- The risk of business failure decreases.
- As the business becomes more attractive to lenders, debt financing may be available at reasonable terms. The company may also have assets that it can use to secure debt.
- However, most companies use debt conservatively to retain their operational and financial flexibility.
- Equity is generally the main source of capital.

Mature Businesses

- In this stage a company typically experiences a slowdown in revenue growth; and growth-related investment spending decreases.
- Cash flow is usually positive and predictable.
- The risk of business failure is low.
- Debt financing is available at attractive terms often on an unsecured basis.
- To take advantage of the cheaper debt (as compared to equity) companies typically use significant leverage.
- Over time a company may experience de-leveraging due to continuous positive cash flow generation and share price appreciation.
- To offset this de-leveraging companies typically buy back shares and reduce the proportion of equity in the capital structure.
- Share buy backs are preferred over cash dividends as they are more tax-efficient and do not set future expectations.
- Investors generally respond favorably to share buyback announcement, which may lead to an increase in share price.

In the section above we established a general relationship between company maturity and capital structure. However, there are two important exceptions:

Capital-Intensive Businesses

Business such as real estate, utilities, shipping, airline etc. are highly capital intensive. Also, the underlying assets can be bought and sold fairly easily and make for a good collateral. Such businesses tend to use high levels of leverage irrespective of their maturity stage.

Capital-Light Businesses

Some businesses such as software can scale easily and do not require substantial

investments in fixed or working capital to support growth. They are typically cash flow positive from an early stage and never need to raise large amounts of capital. Therefore, they tend to use very little debt in their capital structure.

Determinants of the Costs of Debt and Equity

The costs of debt and equity are determined in the financial markets by top-down factors that affect the overall market, as well as investor's assessment of issuer-specific factors.

Top-Down Factors

A company's cost of debt is equal to a benchmark risk-free rate plus a credit spread specific to the company. Market conditions/ business cycle affect both the benchmark rates and the credit spreads. The credit spreads tend to widen during recessions and tighten during expansions.

Companies may increase their use of debt when borrowing is less expensive due to low benchmark interest rates and/or tight credit spreads, and vice versa.

Issuer-Specific Factors

Investors consider the risk and return profile of an issuer and adjust their required rates of return relative to a base rate by evaluating the following factors:

- Sales risk: Some companies like Vodafone in the telecom industry, have very stable revenue streams because a large proportion of their revenues are subscription-like, recurring revenues. This is generally viewed as a positive. In contrast, companies in cyclical industries, such as Toyota in the automobile industry, have more volatile revenue streams that are highly sensitive to fluctuations in the business cycle. This is generally viewed as a negative.
- Profitability risk: Aside from revenue stability, profit margin stability is an important factor, which is determined by a company's proportion of fixed versus variable costs. Operating leverage = $\text{Fixed costs} / \text{Total costs}$. Companies with higher operating leverage experience a greater change in cash flow and profitability for a given change in revenue than firms with low operating leverage
- Financial leverage and interest coverage: The existing financial leverage of a firm influences its capital structure decisions. Highly leveraged firms face a higher risk of default and, as a result, have less capacity to service additional debt. Underleveraged firms, on the other hand, can support additional debt relatively easily.
- Collateral/type of assets owned by the firm: In general, assets that support increased debt use include those that are considered strong collateral, generate cash, or are fungible or liquid, such as real estate, automobiles, aircraft, and receivables from creditworthy customers.

4. Modigliani–Miller Capital Structure Propositions

Economists Modigliani and Miller claimed that given certain assumptions, a company's choice of capital structure does not affect its value.

The assumptions made were:

1. Homogeneous expectations: All investors have the same expectations with respect to the cash flows from an investment in bonds or stock.
2. Perfect capital markets: There are no transaction costs, no taxes, no bankruptcy costs, and everyone has the same information.
3. Risk-free rate: Investors can borrow and lend at the risk-free rate.
4. No agency costs: Managers always act to maximize shareholder wealth.
5. Independent decisions: Financing and investment decisions are independent of each other. Operating income is unaffected by changes in the capital structure.

Modigliani and Miller's work provides us a starting point and allows us to examine what happens when the assumptions are relaxed to reflect real-world considerations.

Capital Structure Irrelevance (MM Proposition I without Taxes)

MM Proposition I: The market value of the company is not affected by the capital structure of the company. This is because individual investors can create any capital structure they prefer for the company by borrowing and lending in their own accounts in addition to holding shares in the company.

$$V_L = V_U$$

where:

V_L = value of the levered firm

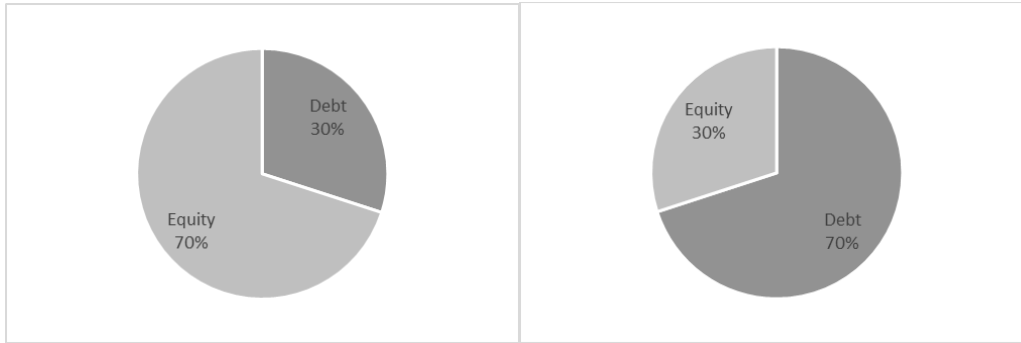
V_U = value of the unlevered firm

In other words, the value of a company is determined solely by its cash flows, not by the relative reliance on debt and equity capital.

The weighted average cost of capital (WACC) is unaffected by the capital structure.

$$\text{where: } WACC = w_d r_d + w_p r_p + w_e r_e$$

We can explain MM's capital structure irrelevance proposition in terms of a pie. Suppose the total value of a company is represented by a pie. Slices represents how much of the total capital is contributed by debt and equity. Depending on the capital structure, the pie can be split in any number of ways, but the size of the pie will remain the same. This is illustrated in the figures below.



Higher Financial Leverage Raises the Cost of Equity (MM Proposition II without Taxes)

MM Proposition II: The cost of equity is a linear function of the company's debt-to-equity ratio.

$$r_e = r_0 + (r_0 - r_d) \frac{D}{E}$$

where:

r_0 is the cost of capital for a company financed only by equity and has zero debt

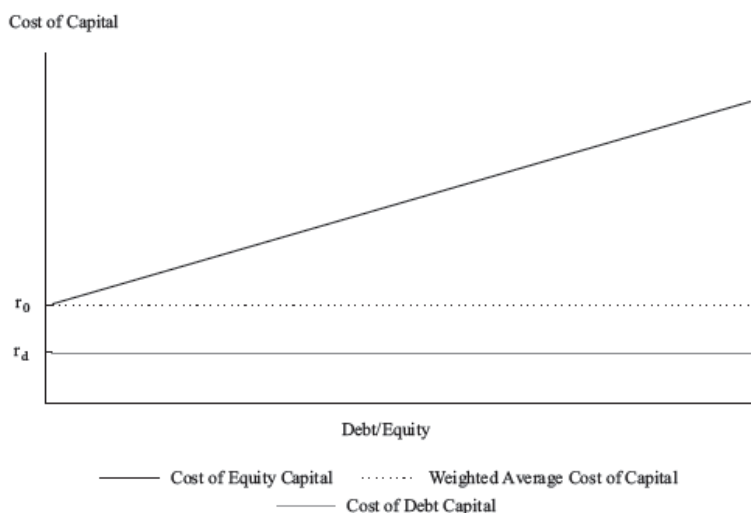
r_d is the cost of debt

r_e is the cost of equity

D/E = debt-to-equity ratio

As D/E rises, i.e. the company increases the use of debt, the cost of equity (r_e) rises. We know from MM Proposition I that the value of a company is unaffected by changes in D/E and the WACC remains constant. Proposition II then implies that the cost of equity increases in such a manner as to exactly offset the increased use of cheaper debt in order to maintain a constant WACC. Under this proposition, WACC is determined by the business risk of the company, and not by the capital structure.

MM Proposition II is illustrated in the figure below.



As leverage increases, the cost of equity increases, but WACC and the cost of debt remain constant.

Firm Value with Taxes (MM Proposition II with Taxes)

The discussion so far has ignored taxes. Now, we will present MM propositions I and II with taxes. As interest paid is tax deductible, the use of debt provides a tax shield that increases the value of a company. If we ignore the costs of financial distress and bankruptcy, the value of the company increases as we take on more debt.

MM Proposition I with taxes: The value of a levered company is equal to the value of an unlevered company plus the value of the debt tax shield.

$$V_L = V_U + tD$$

where:

V_L = value of the levered firm

V_U = value of the unlevered firm

t = marginal tax rate

D = value of debt in the capital structure

The term tD is often referred to as the debt tax shield

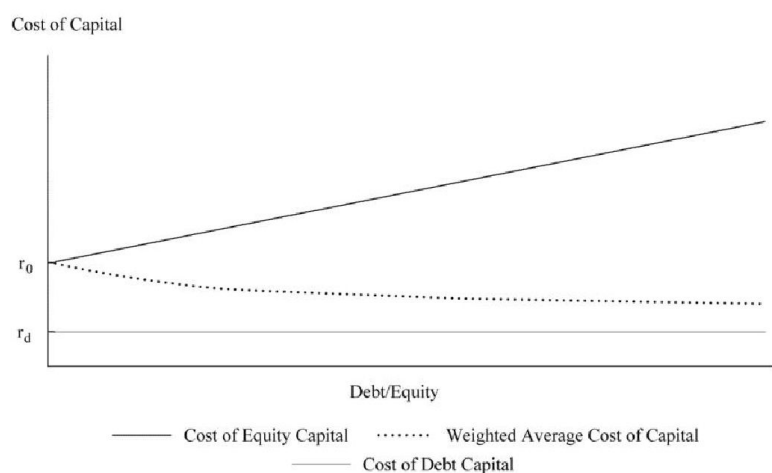
Cost of Capital (MM Proposition II with Taxes)

MM Proposition II with taxes: The cost of equity is a linear function of the company's debt-to-equity ratio with an adjustment for the tax rate.

The cost of equity increases as the company increases the amount of debt in its capital structure, but the cost of equity does not rise as fast as it does in the no tax case.

$$r_e = r_0 + (r_0 - r_d)(1 - t) \frac{D}{E}$$

WACC for a leveraged company falls as debt increases, and therefore the overall company value increases. This is illustrated in the figure below:



This proposition implies that in the presence of taxes (but no financial distress or bankruptcy costs), the use of debt is value enhancing and, at the extreme, 100% debt is optimal.

The table below provides a summary of MM propositions.

	Without Taxes	With Taxes
Proposition I	$V_L = V_U$	$V_L = V_U + tD$
Proposition II	$r_e = r_0 + (r_0 - r_d) D/E$	$r_e = r_0 + (r_0 - r_d)(1 - t) D/E$

Cost of Financial Distress

The disadvantage of operating and financial leverage is that earnings are magnified downward during economic slowdown. Lower or negative earnings put companies under stress, and this financial distress adds costs to a company.

The costs of financial distress can be classified into direct and indirect costs. Direct costs include the actual cash expenses associated with the bankruptcy process, such as legal and administrative fees. Indirect costs include forgone investment opportunities, impaired ability to conduct business, and agency costs associated with the debt during periods in which the company is near or in bankruptcy.

The costs of financial distress are lower for companies whose assets have a ready secondary market. For example, airlines, shipping companies etc.

The probability of financial distress and bankruptcy increases as the degree of leverage increases. The probability of bankruptcy depends, in part, on the company's business risk. Other factors that affect the likelihood of bankruptcy include the company's corporate governance structure and the management of the company.

5. Optimal Capital Structure

So far in our discussion of the MM propositions, we have only relaxed the assumption of no corporate taxes. We now consider a scenario that includes both corporate taxes and the costs of bankruptcy/financial distress.

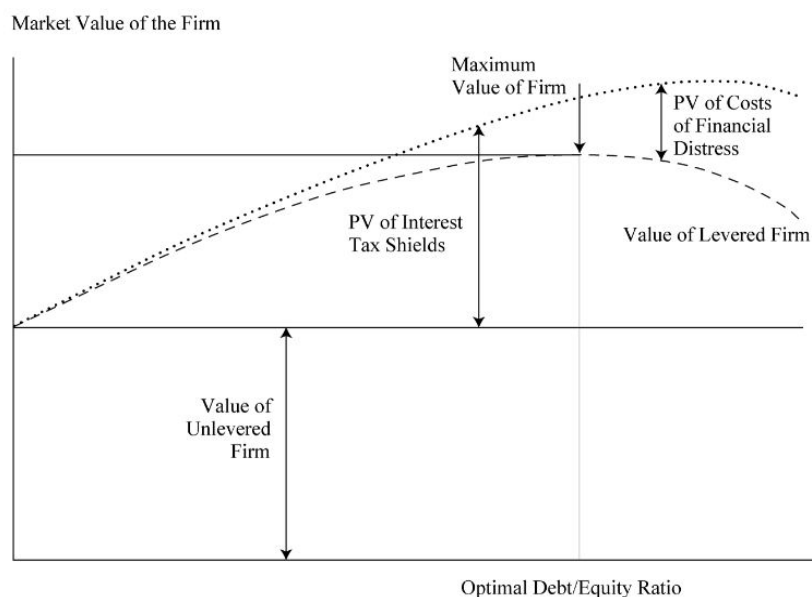
The optimal capital structure is the one at which the value of the company is maximized. The static trade-off theory is based on balancing the expected costs from financial distress against the tax benefits of debt service payments. Considering both the tax shield provided by debt and the costs of financial distress, the expression for the value of a leveraged company becomes:

$$V_L = V_U + tD - PV(\text{Costs of financial distress})$$

This equation represents the static trade-off theory of capital structure. It results in an optimal capital structure where debt is less than 100%. At low levels of debt, the tax benefits of debt outweigh the PV of financial distress, and the firm's value increases. However, as more debt is added, the PV of financial distress begins to rise. At the optimal

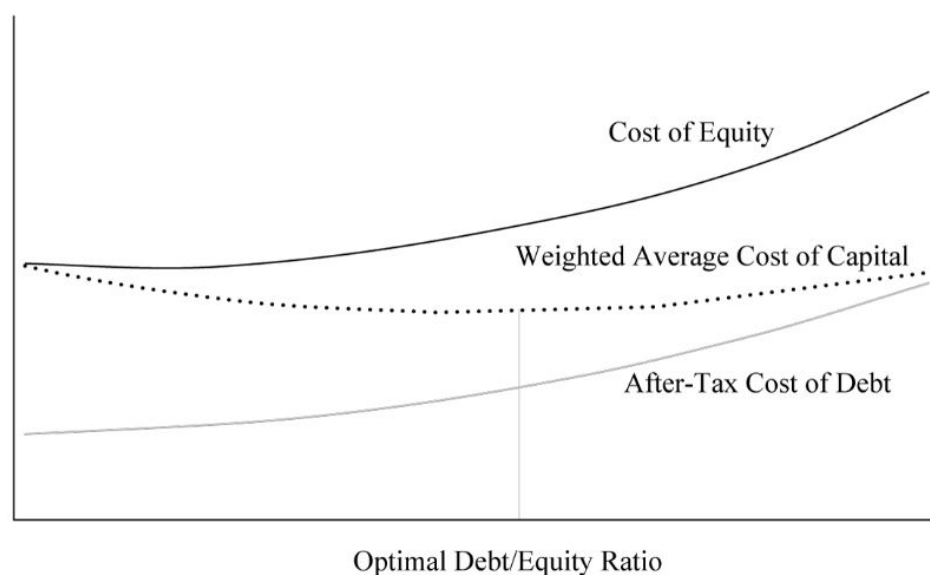
debt/equity ratio point, the PV of financial distress is equal to the tax benefit of debt, and the firm value is maximum. If more debt is added beyond this point, the PV of financial distress outweighs the tax benefits of debt, and the firm's value decreases.

These concepts are illustrated in Exhibit 10 from the curriculum.



WACC is minimum at the point where the firm value is maximized.

Cost of Capital



When a company identifies its most appropriate capital structure, it may adopt this as its target capital structure. However, a company's capital structure may vary from its target

because management may try to take advantage of short-term opportunities in alternate financing sources. Market value variations also continuously affect the company's capital structure. Sometimes, it may be impractical and expensive for a company to maintain its target structure.

In practice, it is difficult to precisely determine the optimal capital structure, because of the difficulty in estimating some costs, such as the costs of financial distress. Therefore, managers often use an optimal capital structure range instead of a precise ratio. For example, instead of saying exactly 40% debt is optimal, managers can say debt should be in the range of 30% to 50%.

Market Value vs. Book Value

The optimal capital structure should be calculated using the market value of equity and debt. However, company capital structure targets often use book values instead because:

- Market values can fluctuate a lot and they do not necessarily impact the appropriate level of borrowing. For example, a company that has seen a rapid share price increase may decide to take advantage of this situation and issue even more equity instead of debt.
- The management is primarily concerned with the amount and types of capital invested 'by' the company and not 'in' the company. Their perspective is different from investors who have purchased securities at the prevailing market price.
- Lenders and rating agencies typically focus on the book value of debt and equity for their calculation measures.

Financing decisions are often opportunistic. Managers consider the share price of their company as well as market interest rates for their debt when deciding when, how much, and what type of capital to raise.

Target Weights and WACC

When conducting an analysis if we know the company's target capital structure, then we should use it in our analysis. However, analysts typically do not know a company's target capital structure. It can be estimated using one of these methods:

1. Assume the company's current capital structure, at market value weights for the components, represents the company's target capital structure.
2. Examine trends in the company's capital structure or statements by management regarding capital structure policy to infer the target capital structure.
3. Use averages of comparable companies' capital structures as the target capital structure.

Example: Estimating the Proportions of Capital

(This is based on knowledge check example from the curriculum.)

The following information is provided for a company:

Market value of debt: EUR50 million

Market value of equity: EUR60 million

Primary competitors and their capital structures (in millions):

Competitor	Market Value of Debt	Market Value of Equity
A	EUR25	EUR50
B	EUR101	EUR190
C	GBP40	GBP60

Calculate the proportions of debt and equity, if the target capital structure is calculated based on:

1. The current capital structure
2. Competitor's capital structure
3. The company's announcement that a debt-to-equity ratio of 0.7 reflects its target capital structure.

Solution to 1:

$$w_d = \frac{50}{50 + 60} = 0.4545$$

$$w_e = \frac{60}{50 + 60} = 0.5454$$

Solution to 2:

$$w_d = \frac{\left(\frac{25}{25+50}\right) + \left(\frac{101}{101+190}\right) + \left(\frac{40}{40+60}\right)}{3} = 0.3601$$

$$w_e = \frac{\left(\frac{50}{25+50}\right) + \left(\frac{190}{101+190}\right) + \left(\frac{60}{40+60}\right)}{3} = 0.6399$$

Solution to 3:

A debt-to-equity ratio of 0.7 represents a weight on debt of $0.7/1.7 = 0.4118$, so $w_d = 0.4118$ and $w_e = 1 - 0.4118 = 0.5882$.

Pecking Order Theory and Agency Costs

Managers have more information about a company's performance and prospects than outsiders, such as owners and investors. This is referred to as information asymmetry. Investors demand higher returns when asymmetry of information is high because this increases the probability of agency costs.

Investors closely watch manager's decision for insights into the company's future prospects. Managers may provide information to investors ("signaling") through their choice of

financing method. Fixed payment commitments, for example, can indicate management's confidence in the company's future prospects.

Being aware of this, managers take into account how their actions might be interpreted by outsiders.

Pecking Order Theory: This theory suggests that managers choose methods of financing that send the least signal to outsiders. The preferred hierarchy for financing is:

- First preference: internal financing (retained earnings)
- Second preference: debt financing
- Third preference: equity financing

Agency Costs: Agency costs are incremental costs that arise from conflicts of interest between managers, shareholders, and bondholders. Agency theory predicts that as the use of debt increases, agency costs to equity will decrease. The more financially leveraged a company is, the less freedom managers have to incur additional debt or waste cash in inefficient ways. This also serves as the basis for the 'free cash flow hypothesis.'

Free Cash Flow Hypothesis: As per this hypothesis, high debt levels discipline managers by forcing them to manage the company efficiently so that the company can make its interest and principal payments. Thus, by reducing the company's free cash flows, managers have fewer opportunities to misuse cash.

Summary

LO: Calculate and interpret the weighted-average cost of capital for a company.

An issuer's cost of capital is estimated using a weighted average of the costs of debt and equity, using either the current market value or management's target weights of each type of financing as the weights.

$$WACC = w_d r_d (1 - t) + w_p r_p + w_e r_e$$

WACC represents the overall cost of capital for the firm and is the appropriate discount rate to use for projects having a similar risk profile as that of the firm.

LO: Explain factors affecting capital structure and the weighted-average cost of capital.

Capital structure refers to the specific mix of debt and equity a company uses to finance its assets and operations. Issuers desire a capital structure that minimizes their weighted-average cost of capital and generally matches the duration of their assets.

The total amount and type of financing needed usually depends on the issuer's business model and its position in the corporate life cycle. As a company matures and progresses from start-up to growth to maturity, its business risk typically decreases, and its operating cash flows turn positive and more predictable. This enables greater use of leverage at more favorable terms. Capital intensive businesses tend to use high levels of leverage irrespective of their maturity stage. Whereas, capital light businesses tend to use very little debt in their capital structure.

The component cost of debt and equity are impacted by top-down factors such as financial market and industry conditions; and by issuer specific factors such as stability of revenues and profit margins, and operating and financial leverage.

LO: Explain the Modigliani–Miller propositions regarding capital structure.

	Without Taxes	With Taxes
Proposition I	$V_L = V_U$	$V_L = V_U + tD$
Proposition II	$r_e = r_0 + (r_0 - r_d) D/E$	$r_e = r_0 + (r_0 - r_d)(1 - t) D/E$

LO: Describe optimal and target capital structures.

The optimal capital structure is the one at which the value of the company is maximized. The static trade-off theory is based on balancing the expected costs from financial distress against the tax benefits of debt service payments. Considering both the tax shield provided by debt and the costs of financial distress, the expression for the value of a leveraged company becomes:

$$V_L = V_U + tD - PV(\text{Costs of financial distress})$$

At the optimal debt/equity ratio point, the PV of financial distress is equal to the tax benefit

of debt, and the firm value is maximum.